

21th September 2026, Analysis of Several Variables Notes,
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Covered: Riemann Integration on $\mathbb{R}^n$

A rigorous theory of integration on $\mathbb{R}^n$ is motivated by many problems in physics, take for instance the use of the density function in deriving Mass/Traffic Flow/Population etc.

$$M = \int_B \rho(x_1, x_2, x_3) \, dx_1 \, dx_2 \, dx_3$$

The aim of this section would be to derive an integration theory for $f : Q \rightarrow \mathbb{R}$ where f is bounded and Q is a rectangle:

$$Q = \prod_{i=1}^m [a_i, b_i] \subseteq \mathbb{R}^m, \quad a_i \leq b_i.$$

Definition 0.1 (Basic Riemann Integral). We consider a bounded function on a closed interval $f : [a, b] \rightarrow \mathbb{R}$. We can partition the interval as $P = \{a = x_0 < x_1 < \dots < x_m = b\}$. We define the sums,

$$\begin{aligned} \text{lower sum} &= \sum_{i=1}^m m_i(x_i - x_{i-1}), & m_i &= \inf_{x \in [x_{i-1}, x_i]} f(x) \\ \text{upper sum} &= \sum_{i=1}^m M_i(x_i - x_{i-1}), & M_i &= \sup_{x \in [x_{i-1}, x_i]} f(x) \end{aligned}$$

The upper and lower Riemann integrals are subsequently defined,

$$\underline{\int} f = \sup(\text{lower sums}), \quad \overline{\int} f = \inf(\text{upper sums}).$$

And finally, *if they are equal*, the *Riemann Integral* is defined

$$\int f = \underline{\int} f = \overline{\int} f$$

1 Riemann Integral on $\mathbb{R}^n$

Consider a bounded rectangle $Q \subseteq \mathbb{R}^m$. A partition of Q is a tuple $P = (P_1, P_2, \dots, P_m)$ where P_i is a partition of $[a_i, b_i]$:

$$P_i = \{a_i = t_0^i < t_1^i < \dots < t_{m_i}^i = b_i\}.$$

A rectangle R is a *subrectangle* of Q iff

$$R = \prod_{i=1}^m [t_{j_i-1}^i, t_{j_i}^i],$$

where $[t_{j_i-1}^i, t_{j_i}^i]$ is a subinterval given by P_i .

Definition 1.1. We define the volume of a rectangle Q as

$$\text{vol } Q := \prod_i (b_i - a_i),$$

Theorem 1.2. $\text{vol } Q = \sum_{R \text{ subrectangle}} \text{vol } R$.

Proof.

$$\begin{aligned} \text{vol } Q &= \prod_i (b_i - a_i) \\ &= \prod_i \left(\sum_{j=1}^{m_i} (t_j^i - t_{j-1}^i) \right) \\ &= \sum_{j=1}^{m_i} \prod_i (t_j^i - t_{j-1}^i) = \sum_R \text{vol } R. \end{aligned}$$

□

Let $f : Q \rightarrow \mathbb{R}$ be a bounded function. Given a subrectangle R , we define

$$m_R(f) := \inf_{x \in R} f(x), \quad M_R(f) := \sup_{x \in R} f(x).$$

The lower sum and upper sums are analogous to the basic Riemann sums case

$$\begin{aligned} L(f, P) &= \sum_{\substack{R \\ \text{subrectangle}}} m_R(f) \text{vol}(R) \\ U(f, P) &= \sum_{\substack{R \\ \text{subrectangle}}} M_R(f) \text{vol}(R) \end{aligned}$$

Few things are immediate.

1. $L(f, P) \leq U(f, P)$ since $m \leq M$.
2. There exist $m, M \in \mathbb{R}$ such that

$$m \leq L(f, P) \leq U(f, P) \leq M.$$

Or, $\exists m_0$ such that $m_0 \leq m_R(f)$ for all R with $m_0 = \inf_{x \in Q} f(x)$. Similarly, $\exists M_0$ such that $M_R(f) \leq M_0$ for all R with $M_0 = \sup_{x \in Q} f(x)$

Hence

$$\begin{aligned}
m_0 \operatorname{vol}(Q) &= \sum_R m_0 \operatorname{vol}(R) \\
&\leq \sum_R m_R(f) \operatorname{vol}(R) \\
&\leq \sum_R M_R(f) \operatorname{vol}(R) \leq M_0 \operatorname{vol}(Q).
\end{aligned}$$

Definition 1.3. P' is a *refinement* of P iff $P'_i \supseteq P_i \forall i$

Theorem 1.4. Moreover, given P_1, P_2 partitions of Q , we have $L(f, P_1) \leq U(f, P_2)$

Proof. If P_1, P_2 are partitions of Q , then $P_1 \cup P_2$ is a refinement of both P_1 and P_2 . Therefore,

$$L(f, P_1) \leq L(f, P_1 \cup P_2) \leq U(f, P_1 \cup P_2) \leq U(f, P_2).$$

□

Theorem 1.5. If P' is a refinement of P , then

$$L(f, P) \leq L(f, P').$$

Proof. It is enough to show that introducing a single new partition point cannot decrease the lower sum. Suppose P' is obtained from P by adding one point t to the k -th coordinate partition. Let $R = \prod_{i=1}^m [a_i, b_i]$ be a subrectangle of P which is split by this new point. Then

$$R = R_1 \cup R_2, \quad \operatorname{vol}(R) = \operatorname{vol}(R_1) + \operatorname{vol}(R_2),$$

and since $R_1, R_2 \subseteq R$,

$$m_R(f) \leq m_{R_1}(f), \quad m_R(f) \leq m_{R_2}(f).$$

Therefore,

$$\begin{aligned}
m_R(f) \operatorname{vol}(R) &= m_R(f) (\operatorname{vol}(R_1) + \operatorname{vol}(R_2)) \\
&\leq m_{R_1}(f) \operatorname{vol}(R_1) + m_{R_2}(f) \operatorname{vol}(R_2).
\end{aligned}$$

All subrectangles not affected by the new point contribute exactly the same amount to both lower sums. Hence

$$L(f, P) \leq L(f, P').$$

Now a refinement P' is obtained from P by introducing finitely many new partition points. Repeating the above argument for each new point gives

$$L(f, P) \leq L(f, P').$$

□

Definition 1.6. We can finally define the upper and lower integrals for this setup.

$$\underline{\int} f := \sup_P L(f, P), \quad \overline{\int} f := \inf_P U(f, P).$$

Theorem 1.7.

$$\underline{\int} f \leq \overline{\int} f.$$

Proof. For any partitions P_1, P_2 , by previous theorem we have,

$$L(f, P_1) \leq U(f, P_2) \quad \forall P_1, P_2.$$

Taking the supremum over P_1 ,

$$\underline{\int} f = \sup_{P_1} L(f, P_1) \leq U(f, P_2) \quad \forall P_2.$$

Taking the infimum over P_2 ,

$$\underline{\int} f \leq \inf_{P_2} U(f, P_2) = \overline{\int} f.$$

□

Yay!